

Summary Data on Acid Cropland and Lime Requirements in Ghana

GAIA

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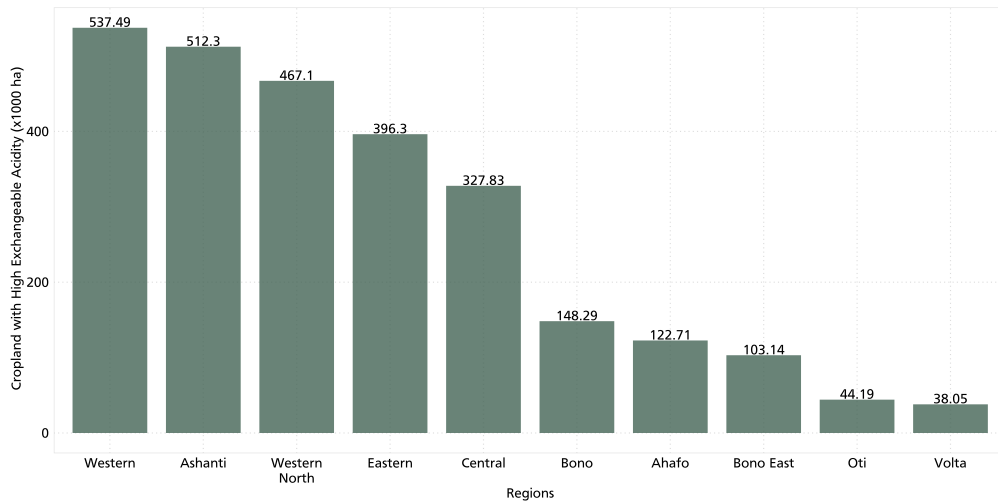
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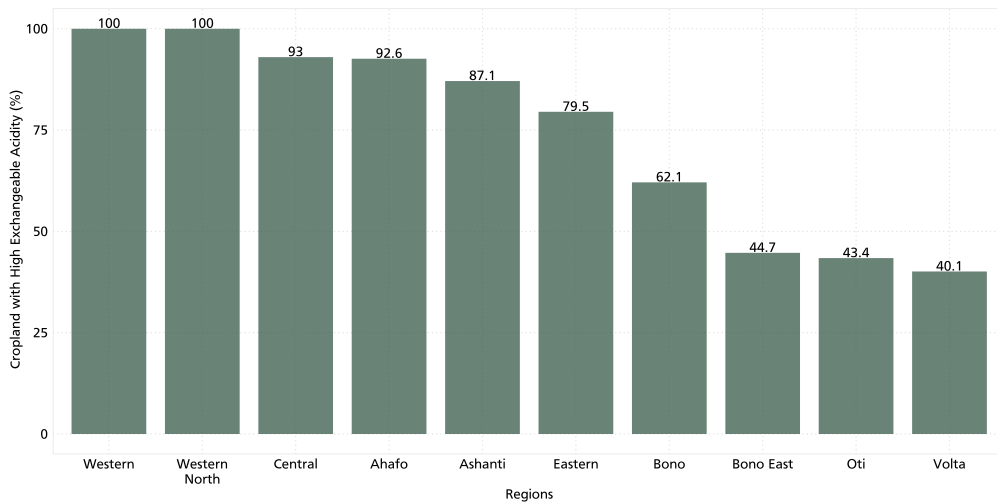
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Ghana**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production, ROI and profitability from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

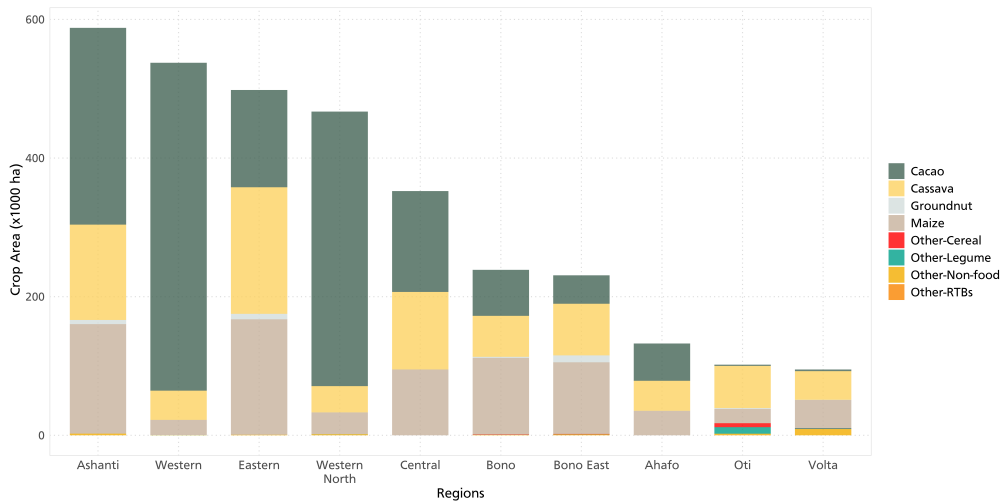
Top 10 Regions in Ghana with the highest acidic cropland



Top 10 Regions in Ghana with the highest acidic cropland (Percentage of total cropland)



Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland

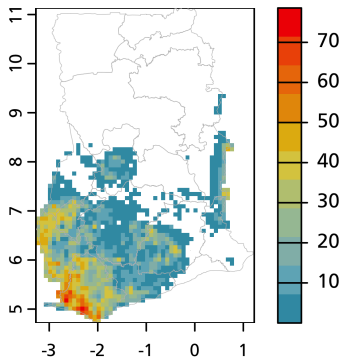


Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland (table)

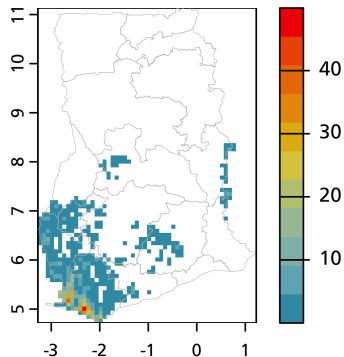
Regions	Cacao	Cassava	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs
Ahafo	53.81	43.35	0.05	35.19	-	-	0.09	-
Ashanti	283.89	137.56	6.01	157.99	0.01	-	2.39	-
Bono	66.38	59.12	1.56	110.49	0.49	0.06	0.66	0.03
Bono East	41.02	74.51	9.93	103.40	0.47	0.24	1.28	-
Central	145.51	111.92	-	94.94	-	-	0.02	-
Eastern	140.31	182.65	7.89	166.59	-	-	0.77	-
Oti	1.80	60.83	1.24	20.55	5.71	9.47	2.33	-
Volta	2.37	41.34	0.26	40.49	0.37	1.09	9.03	0.01
Western	473.00	42.19	0.05	21.83	-	0.02	0.39	0.01
Western North	396.12	37.88	0.16	31.23	0.01	0.08	1.56	0.05

Yield loss due to soil acidity for Maize and Cacao in Ghana

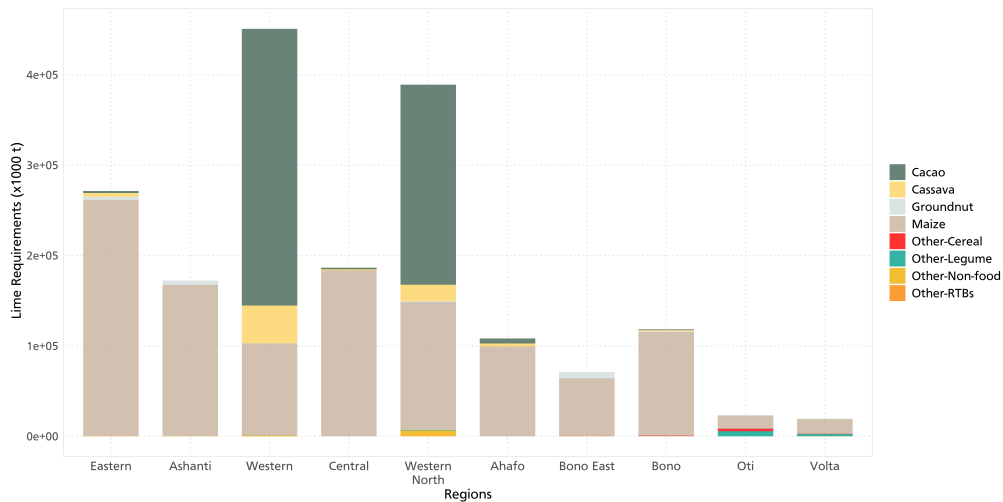
Maize Yield Loss (%)



Cacao Yield Loss (%)



Lime requirements (t) for acid soil remediation in Ghana

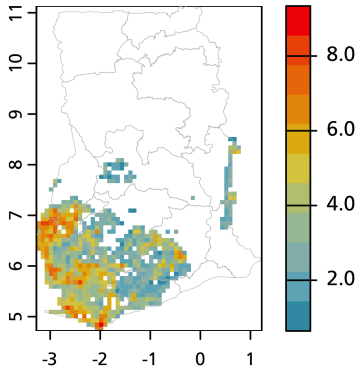


Lime requirements (1000 t) for acid soil remediation in Ghana (table)

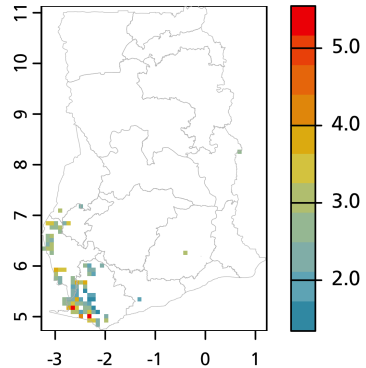
Regions	Cacao	Cassava	Groundnut	Maize	Other- Non-food	Other- Cereal	Other- Legume	Other- RTBs
Ahafo	5.58	3.26	0.03	99.39	0.03	-	-	-
Ashanti	-	-	4.89	167.19	0.28	-	-	-
Bono	0.90	1.27	0.66	114.87	0.20	0.34	0.12	0.04
Bono East	-	-	6.95	63.76	0.31	0.14	-	-
Central	1.80	1.20	-	183.57	0.02	-	-	-
Eastern	2.22	4.15	3.47	261.25	0.37	-	-	-
Oti	-	-	0.72	14.34	0.05	2.74	5.66	-
Volta	-	-	0.36	16.26	0.10	0.47	2.28	-
Western	306.18	41.70	0.29	101.58	0.98	-	0.10	0.04
Western North	221.42	18.12	1.14	141.47	6.09	0.03	0.56	0.30

Lime rate for Maize and Cacao in Ghana

Maize Lime Rate (t/ha)

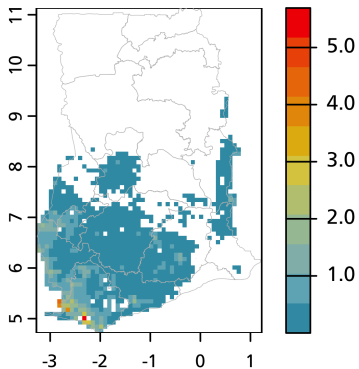


Cacao Lime Rate (t/ha)

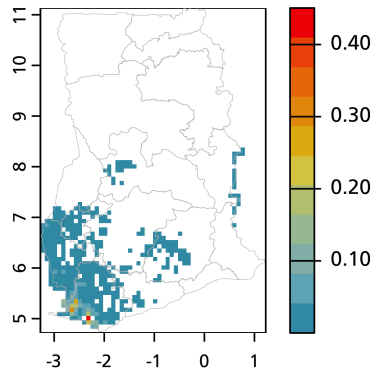


Yield response to lime application in Ghana

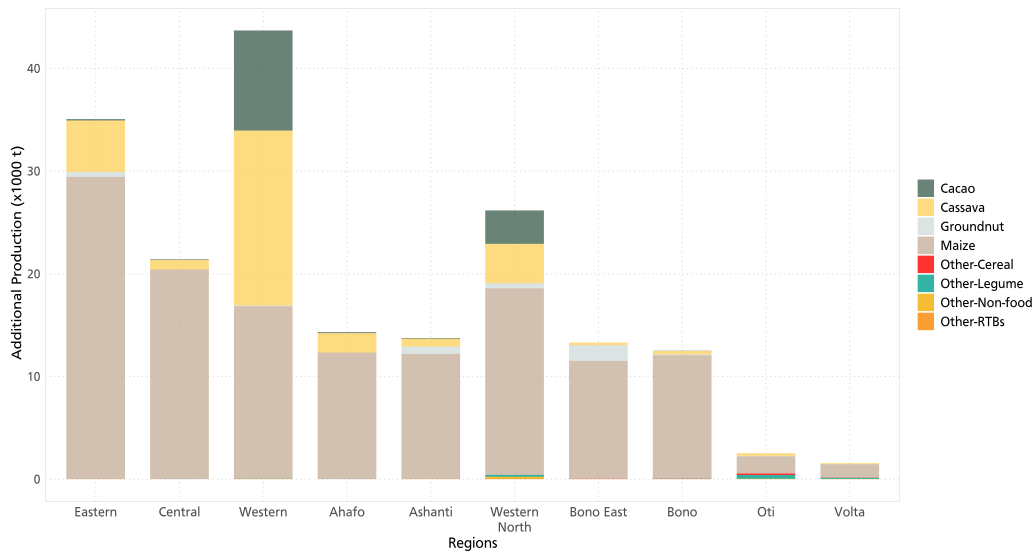
Maize Yield Response (t/ha)



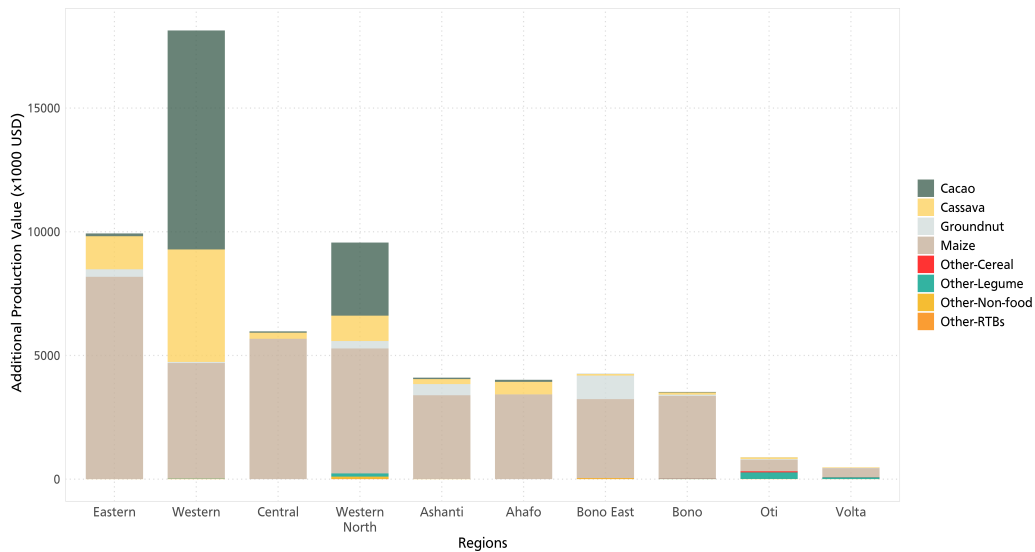
Cacao Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Regions in Ghana



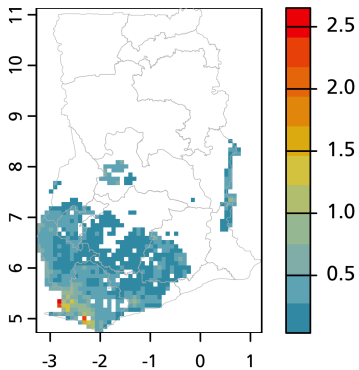
Potential for additional production value (US\$) in top 10 acidic Regions in Ghana



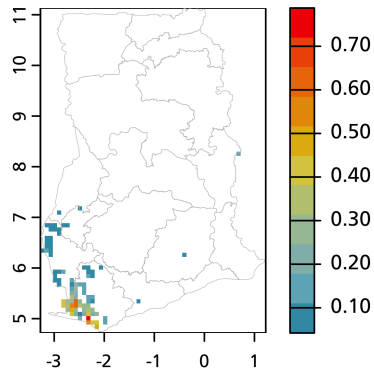
- ROI for Lime Application
- Return on investment for lime application in Ghana

Return on investment for lime application in Ghana

Maize ROI (USD/USD)

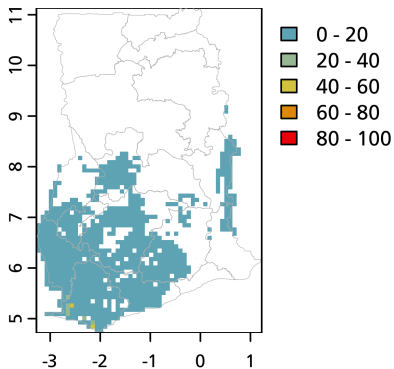


Cacao ROI (USD/USD)

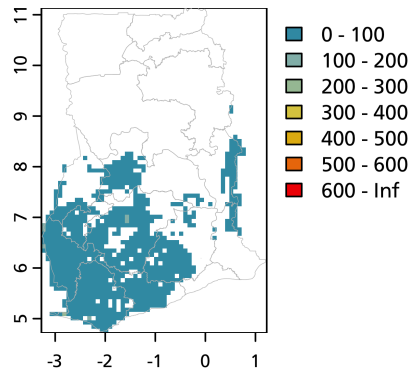


Profitability of lime application in Ghana

Profitable area (%)



Weighted profit (USD/ha)



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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